11 Use Cases

Use Case Requirements Overview

هيئة كهرباء ومياه دبي
Dubai Electricity & Water Authority

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Use Case 01

AI Assisted Personalized Learning Paths

Current State

Teachers create static learning paths and adjust pacing after periodic assessments.

AI Transformation

Continuous learner models synthesize assessment results, engagement signals, prior performance, and competency maps to generate dynamic, individualized learning paths.

The AI adapts content sequencing, recommends remediation or acceleration, and proposes micro-learning activities in real time. Recommendations include rationale and confidence scores and are surfaced to teachers for review or override.

Key Benefits

- Scalable personalization
- Faster mastery
- Reduced teacher prep
- Targeted remediation

Primary Users

Students | Teachers | Academic Coaches

Success Metrics

- Mastery rate improvement
- Reduction in time-to-competency
- Teacher time saved

Use Case 02

AI Parent Insight and Progress Reports

Current State

Parents receive periodic static reports (attendance, grades).

AI Transformation

Automated narrative generation synthesizes multi-source signals into concise, actionable progress briefs: trend summaries, risk flags, suggested home activities, and short coaching scripts for parents.

Reports are personalized by child, translated where needed, and scheduled or pushed on demand. Each insight includes the AI's reasoning and suggested next steps.

Key Benefits

- Higher parent engagement
- Earlier interventions
- Clearer home-school collaboration

Primary Users

Parents | Class Teachers

Success Metrics

- Parent engagement rate
- Follow-up action rate
- Satisfaction scores

Use Case 03

Autonomous Timetable & Class Scheduling

03

Current State

Timetables built manually or with legacy tools; many iterations to resolve conflicts.

AI Transformation

An AI scheduler ingests constraints (staff skills, room capacity, curriculum rules, KHDA/BTEC constraints) and generates optimized, conflict-free timetables.

It continuously re-optimizes in response to absences, enrollment changes, or resource failures and explains tradeoffs (e.g., which constraint was relaxed and why). Human overrides are tracked and fed back to improve models.

Key Benefits

- Faster publishing
- Fewer conflicts
- Better utilization
- Rapid adaptation to change

Primary Users

Timetable Admins

School Leadership

Success Metrics

- Time to publish
- Post-publish conflict rate
- Utilization metrics

Use Case 04

AI Assisted Homework and Assessment Generation

Current State

Teachers author assignments and quizzes manually; quality and alignment vary.

AI Transformation

Generative models produce differentiated assignments, question variants, rubrics, and scaffolding aligned to learning outcomes and Bloom's taxonomy.

The AI suggests difficulty levels per student, creates auto-gradable items, and provides distractor analysis for MCQs. Teachers edit and approve content; provenance and alignment metadata are stored.

Key Benefits

- Faster content creation
- Consistent standards alignment
- Personalized practice

Primary Users

Teachers

Curriculum Designers

Success Metrics

- Time saved per teacher
- Usage of AI-generated items
- Student performance on generated tasks

Use Case 05

AI Enabled Automated Grading and Feedback

Current State

Manual grading delays feedback and introduces inconsistency.

AI Transformation

Automated scoring handles objective items and rubric-based subjective work; NLP models generate formative feedback and highlight evidence in student submissions.

The system provides confidence scores and routes low-confidence or high-stakes items for human review. Feedback templates are personalized to student learning profiles.

Key Benefits

- Immediate feedback
- Consistent grading
- Reduced teacher workload

Primary Users

Teachers | Students | Examiners

Success Metrics

- Feedback turnaround time
- Inter-rater consistency
- Teacher review time

Use Case 06

Multi-Modal Attendance Fusion

06

Current State

Attendance via manual entry or single devices; reconciliation overhead.

AI Transformation

Fusion models combine facial recognition, RFID/punch logs, mobile check-ins, and teacher marks to produce a reconciled attendance record.

Anomaly detectors flag suspicious patterns (ghost attendance, repeated mismatches) and auto-generate suggested resolutions with confidence and evidence. Attendance triggers for KHDA/BTEC are automated.

Key Benefits

- Accurate records
- Automated compliance
- Fewer manual corrections

Primary Users

Attendance Admins

Compliance Officers

Success Metrics

- Reconciliation workload reduction
- Accuracy vs baseline
- Compliance automation rate

Use Case 07

Student Risk Prediction and Early Intervention

Current State

Risk identification relies on manual review of grades and attendance.

AI Transformation

Predictive models continuously score students for academic, behavioral, and retention risk using multi-source signals. The AI recommends prioritized interventions (tutoring, counseling, parent outreach), generates intervention plans with measurable goals, and tracks outcomes to refine predictions.

Explanations of key risk drivers are provided for transparency, allowing educators to understand the factors contributing to a student's risk profile.

Key Benefits

- Early, targeted support
- Improved retention
- Measurable intervention impact

Primary Users

Counselors | Student Success Teams | Teachers

Success Metrics

- Prediction precision/recall
- Intervention success rate
- Reduction in dropouts

Use Case 08

Conversational AI for Support and Service Requests

Current State

Service desks handle requests manually; slow response times.

AI Transformation

Conversational agents intake requests, interpret intent, pre-fill forms, provide policy-aware answers, and execute routine actions (leave submission, appointment booking).

Complex or sensitive cases are escalated with context and suggested next steps. Conversations are logged and summarized for staff.

Key Benefits

- Faster resolution
- 24/7 basic support
- Reduced admin load

Primary Users

Parents & Students | Staff | Service Desk Teams

Success Metrics

- Bot resolution rate
- Average response time
- Ticket volume reduction

Use Case 09

AI Driven Remote Proctoring and Exam Integrity

09

Current State

Remote exams use basic lockdown tools or manual proctoring.

AI Transformation

Multi-modal proctoring analyzes video, audio, screen activity, and environment signals to detect anomalies (unauthorized persons, device switching, suspicious gaze patterns).

The AI generates incident reports with timestamps, confidence scores, and suggested severity for QA review and appeals.

Key Benefits

- Scalable integrity
- Audit trails
- Reduced cheating

Primary Users

Exam Admins | Invigilators | QA Teams

Success Metrics

- Validated incident rate
- False positive rate
- QA review time

Use Case 10

AI Resource and Staff Allocation Optimization

Current State

Rostering and resource allocation are manual and reactive.

AI Transformation

Forecasting and optimization models recommend staffing levels, substitute assignments, and room allocations based on enrollment forecasts, absence patterns, qualifications, and regulatory constraints.

Key Benefits

- Cost efficiency
- Compliance with ratios
- Improved continuity

Primary Users

HR | Operations | School Leadership

Success Metrics

- Overtime reduction
- Fill rate for shifts
- Compliance adherence

Use Case 11

Adaptive AR/VR Learning Orchestrated by AI

Current State

AR/VR modules are not available.

AI Transformation

AI personalizes immersive scenarios to learner proficiency, adapts difficulty and branching in real time, captures rich interaction telemetry, and maps experiential outcomes to competencies.

The AI recommends next scenarios and integrates assessment checkpoints within the immersive flow.

Key Benefits

- Deep engagement
- Measurable skill transfer
- Scalable personalized immersive learning

Primary Users

Students | Teachers | Instructional Designers

Success Metrics

- Competency gains post-experience
- Engagement duration
- Adoption rate

Thank You

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